

Algebra II with Trigonometry

Chapter 6.2

Olympiad Problems

Saved tutoring response with MathJax rendering. Original PDF file:
[4506907_alg-2trig-h-chp-6-2-note-docx.pdf](#)

Problems

1. Without using a calculator, evaluate $\cos(225^\circ) + \sin(495^\circ)$. Quickly justify your reasoning and sign choices.
2. If $\tan t = -\frac{3}{4}$ and t is in Quadrant II, find the exact value of $\sin t + \cos t$.
3. Find all t in $0 < t < 2\pi$ such that $\sec t < 0$ and $\cot t > 0$. Express your answer in interval form.
4. Let $y = 7 \sin x + 24 \cos x$. What is the maximum value of y as x varies over all real numbers? (No calculus.)
5. Suppose $\sin t + \cos t = 1$. Determine the value of $\sin^4 t + \cos^4 t$.
6. Evaluate $\tan(13\pi) + \cot(17\pi)$ mentally, using periodicity and symmetry.